

CH11.1 平面向量

1. $-\vec{a} + 2\vec{b}$ 2. -1 3. $\left(\frac{13}{36}, \frac{7}{36}\right)$

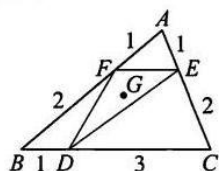
4. $(-2, 3)$ 或 $(-10, 11)$ 5. 10 6. (A)(D) 7. 120

8. (1) $\left(\frac{2}{9}, \frac{1}{3}\right)$ (2) $\frac{9}{5}$ 9. $(25, -15)$ 10. 144

解析

1. $\overrightarrow{AE} = \overrightarrow{AD} + \overrightarrow{DE} = 2\overrightarrow{BC} - \overrightarrow{AB} = \underline{-a + 2b}$

2. $\vec{a} + 2\vec{b} = (1, 2) + (2x, -4) = (2x + 1, -2)$ $2\vec{a} - \vec{b} = (2, 4) - (x, -2) = (2 - x, 6)$
 $\therefore \frac{2x+1}{-2} = \frac{2-x}{6}, 12x + 6 = 2x - 4, x = \underline{-1}$



3. $\because G$ 点 $\triangle DEF$ 的重心

$\therefore$ 對平面上任一點 O ，均有

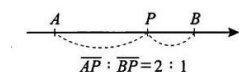
$$\begin{aligned}\overrightarrow{OG} &= \frac{1}{3}(\overrightarrow{OD} + \overrightarrow{OE} + \overrightarrow{OF}) = \frac{1}{3}\left(\frac{3\overrightarrow{OB} + \overrightarrow{OC}}{4} + \frac{2\overrightarrow{OA} + \overrightarrow{OC}}{3} + \frac{2\overrightarrow{OA} + \overrightarrow{OB}}{3}\right) \\ &= \frac{1}{3} \cdot \frac{1}{12}(9\overrightarrow{OB} + 3\overrightarrow{OC} + 8\overrightarrow{OA} + 4\overrightarrow{OC} + 8\overrightarrow{OA} + 4\overrightarrow{OB}) \\ &= \frac{1}{36}(16\overrightarrow{OA} + 13\overrightarrow{OB} + 7\overrightarrow{OC})\end{aligned}$$

取 $O = A$ 得 $\overrightarrow{AG} = \frac{1}{36}(\vec{0} + 13\overrightarrow{AB} + 7\overrightarrow{AC})$

$$= \frac{13}{36}\overrightarrow{AB} + \frac{7}{36}\overrightarrow{AC}$$

$\therefore$ 對數 $(\alpha, \beta) = \left(\frac{13}{36}, \frac{7}{36}\right)$

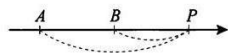
4. 令 $P(x, y)$



(1) 若 P 在 $\overline{AB}$ 上，則如右圖

$$\overrightarrow{AP} = \frac{2}{3}\overrightarrow{AB}$$

$$\Rightarrow (x - 2, y + 1) = \frac{2}{3}(-4 - 2, 5 + 1)$$



$$\overline{AP} : \overline{BP} = 2 : 1$$

$$\Rightarrow (x, y) = (-2, 3)$$

(2) 若 P 不在 $\overline{AB}$ 上，則如右圖

$$\overrightarrow{AP} = 2\overrightarrow{AB}$$

$$\Rightarrow (x - 2, y + 1) = 2(-4 - 2, 5 + 1)$$

$$\Rightarrow (x, y) = \underline{(-10, 11)}$$

$$5. \vec{a} = \overrightarrow{QP} = (8 - (-2), 9 - 4) = (10, 5)$$

$$\vec{b} = \overrightarrow{QR} = (1 - (-2), 8 - 4) = (3, 4)$$

$\vec{a}$ 在 $\vec{b}$ 上之正射影長為

$$\frac{|\vec{a} \cdot \vec{b}|}{|\vec{b}|} = \frac{|10 \times 3 + 5 \times 4|}{\sqrt{9 + 16}}$$

$$= \underline{10}$$

$$6. \text{ 如題圖， } \overrightarrow{OP} = r\overrightarrow{OC} + s\overrightarrow{OB} = 3r\overrightarrow{OA} + s\overrightarrow{OB}$$

當 $r > 0, s > 0$, 且 $r + s > 1$ 時, 終點 P 會落在陰影區域內

$$(A) \because 2\overrightarrow{OA} + \frac{1}{2}\overrightarrow{OB} = \frac{2}{3}(3\overrightarrow{OA}) + \frac{1}{2}\overrightarrow{OB}, \frac{2}{3} + \frac{1}{2} > 1$$

$$(B) x: \frac{3}{2}\overrightarrow{OA} + \frac{1}{3}\overrightarrow{OB} = \frac{1}{2}(3\overrightarrow{OA}) + \frac{1}{3}\overrightarrow{OB}, \frac{1}{2} + \frac{1}{3} < 1$$

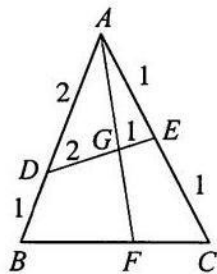
$$(C) \times: 3\overrightarrow{OA} - \frac{1}{2}\overrightarrow{OB} = (3\overrightarrow{OA}) + \left(-\frac{1}{2}\right)\overrightarrow{OB}, -\frac{1}{2} < 0$$

$$(D) \because \frac{9}{4}\overrightarrow{OA} + \frac{1}{3}\overrightarrow{OB} = \frac{3}{4}(3\overrightarrow{OA}) + \frac{1}{3}\overrightarrow{OB}, \frac{3}{4} + \frac{1}{3} > 1$$

$$(E) x: -\frac{3}{5}\overrightarrow{OA} + \frac{9}{5}\overrightarrow{OB} = -\frac{1}{5}(3\overrightarrow{OA}) + \frac{9}{5}\overrightarrow{OB} \quad -\frac{1}{5} < 0 \quad (\text{不合})$$

故選(A)(D)

$$\begin{aligned} 7. \quad |3\vec{a} - 2\vec{b}|^2 &= (3\vec{a} - 2\vec{b}) \cdot (3\vec{a} - 2\vec{b}) = 9|\vec{a}|^2 - 12\vec{a} \cdot \vec{b} + 4|\vec{b}|^2 \Leftrightarrow (6\sqrt{3})^2 = \\ 9 \times 2^2 - 12|\vec{a}||\vec{b}|\cos\theta + 4 \times 3^2 &\Rightarrow 12 \times 2 \times 3 \times \cos\theta = 36 + 36 - 108 \Rightarrow \\ \cos\theta = -\frac{1}{2} \therefore \theta = 120^\circ, &\text{即 } \vec{a} \text{ 與 } \vec{b} \text{ 的夾角為 } 120^\circ \end{aligned}$$



$$8. (1) \overrightarrow{AG} = \frac{\overrightarrow{AD} + 2\overrightarrow{AE}}{3}$$

$$\text{又 } \overrightarrow{AG} = x\overrightarrow{AB} + y\overrightarrow{AC} = x\left(\frac{3}{2}\overrightarrow{AD}\right) + y(2\overrightarrow{AE}) = \frac{3x}{2}\overrightarrow{AD} + 2y\overrightarrow{AE}$$

$$\therefore \frac{3x}{2} = \frac{1}{3}, 2y = \frac{2}{3} \therefore x = \frac{2}{9}, y = \frac{1}{3}$$

故對數 $(x, y) = \left(\frac{2}{9}, \frac{1}{3}\right)$

$$(2)\overrightarrow{AF} = k\overrightarrow{AG} = k\left(\frac{2}{9}\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC}\right)$$

$$\because B、F、C \text{ 共線} \therefore \frac{2k}{9} + \frac{k}{3} = 1 \Rightarrow k = \frac{9}{5}$$

9. 由柯西不等式得

$$(x^2 + y^2)[4^2 + (-3)^2] \geq [x \cdot 4 + y \cdot (-3)]^2$$

$$\Rightarrow 16 \times 25 \geq (4x - 3y)^2 \Rightarrow -20 \leq 4x - 3y \leq 20$$

$$\Leftrightarrow -15 \leq 4x - 3y + 5 \leq 25 \therefore \text{數對 } (M, m) = (25, -15)$$

10. $\vec{a}$ 與 $\vec{b}$ 所張成的平行四邊形 $\square = 3 \times 4 \times \sin 150^\circ = 6$ 則 $2\vec{a} + 4\vec{b}$ 與

$5\vec{a} - 2\vec{b}$ 所張成的平行四邊形面積為 $\left| \begin{vmatrix} 2 & 4 \\ 5 & -2 \end{vmatrix} \right| \times \square = 24 \times 6 = \underline{144}$